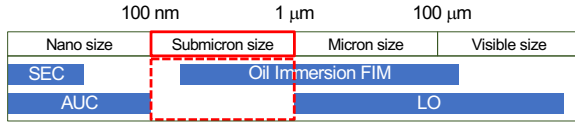


Introduction

Proteins in biopharmaceuticals form aggregates

- Aggregates may cause harmful effects and reduce the effectiveness of the medicine.
- Submicron-sized aggregates (100 nm–1 μ m) have been reported to exhibit higher immunogenicity compared to other size ranges.

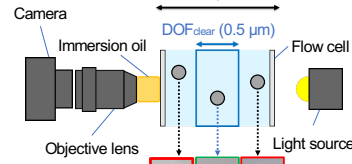


SEC: Size Exclusion Chromatography FIM: Oil Immersion Flow Imaging Microscopy
AUC: Analytical Ultracentrifugation LO: Light Obscuration

- Oil Immersion FIM is a candidate for Submicron size particles analyzing

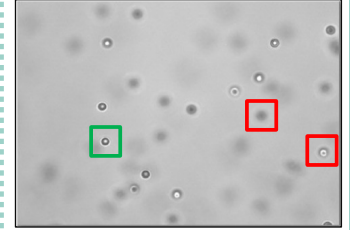
Oil immersion FIM and method development

- Oil immersion FIM is an optical system capturing particle image thus acquire size and morphology information of particle.



$$DoF_{clear} = \frac{n_{medium} \cdot \lambda}{NA^2} + \frac{n_{medium}}{M \cdot NA} e$$

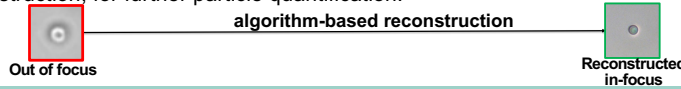
- λ : wavelength
- NA: number of aperture
- n: refractive index
- M: total magnification rate
- e: pixel pitch



- Limited particles can be detected due to narrow depth of field (DOF).
- It is not feasible for analysis of particle concentration and size distribution.

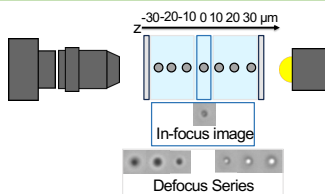
Objective

- develop a computational image-processing method that takes a single image as input and restores defocused particles image to in-focus state through algorithm-based reconstruction, for further particle quantification.

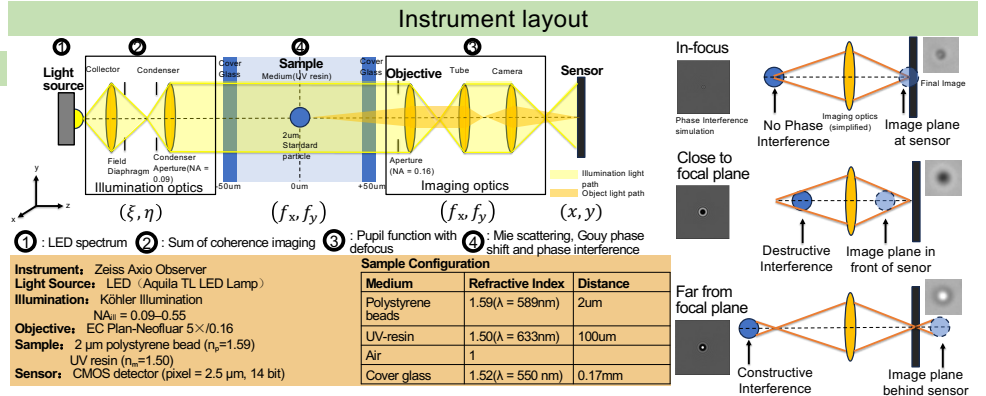


Material & Method

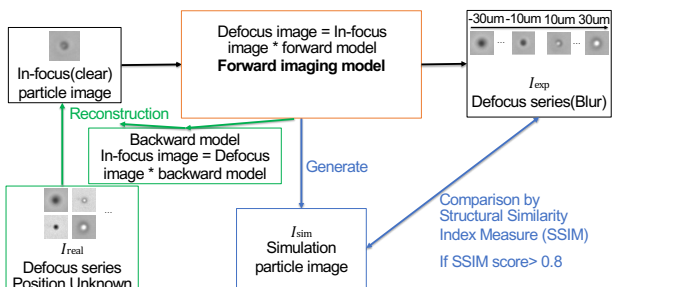
Image acquiring



- Images captured from a Bright Field Microscopy which have the same system setting as Oil Immersion FIM
- Particle immobilized by UV resin. In-focus image and Defocus Series captured from same particle

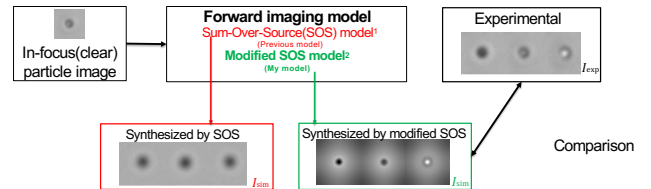


Workflow



- It is essential to development a Forward imaging model for image reconstruction

Forward Imaging model

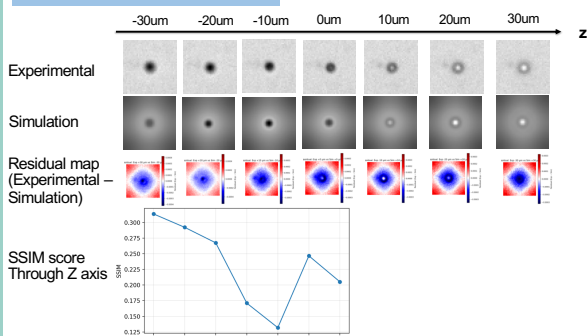


$$I(x, y) = \iint S(\xi, \eta) |F^{-1}[T(f_x - \xi, f_y - \eta)P(f_x, f_y)]|^2 d\xi d\eta \quad (1)$$

$$I(x, y) = \iint S_{LED}(\lambda) d\lambda \iint S_{LED}(\lambda) |F^{-1}[T(f_x, f_y, \lambda, z_{eff})]|^2 d\xi d\eta \quad (2)$$

Legend:
Not feasible for LED illumination case
Not specified for protein aggregates (pure phase specimens): No phase interference
Specified for LED illumination case by spectrum simulation
Phase interference simulation included, performs light behaviors under defocus situation

Result



- Experimental defocus 2um particle images (Experimental) are compared with simulation images (Simulation).
- This discrepancy may cause from different energy distribution between experimental image and simulation image, basically on background.

Summary

- A forward imaging model was built based on an optical system of polychromatic illumination.
- Optical theorem including Mie scattering, Gouy phase shift and phase interference are considered.
- Defocus series of 2um particle was synthesized by forward model and compared with experimental image. Light propagation pattern during defocus was simulated successfully.
- However, low SSIM score presents low similarity, residual map indicates this discrepancy may cause from different energy distribution between experimental image and simulation image, basically on background.

Future Plan

- Forward imaging model requires improving, parameters should be adjusted to compromise the discrepancy.
- Backward imaging model should be derived from forward model.
- Simulation target should be aggregates of biopharmaceutical. Rational size and morphology of aggregates should be investigated and considered into simulation.